EXERCISE 4

SQL: JOINS, UNION, FILTERING and Aggregates Questions

1. INNER JOIN

```
SELECT Employee_ID, First_name, Last_name, Department,
       Salary, Project_id, Project_name, Budget, Status
FROM Employees
INNER JOIN Projects
ON Employees.Employee_ID = Projects.Employee_ID;
```

Employee_ID	first_name	Last_name	Department	Salary	Project_ID	Project_name	Budget	Status
1	John	Doe	IT	70000	101	AI_Development	100 000	Completed
2	Alice	Smith	HR	60000	102	Employee-Training	50000	ongoing
~~1~~	~~Bob~~ John	~~Johnson~~ Doe	~~Finance~~ IT	70000	103	Cyber Security Audit	75000	Pending
~~#~~ 3	~~David~~ Bob	~~BROWN~~ Johnson	~~IT~~ Finance	75000	104	Financial Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65 000	Completed
6	Michael	Clark	Finance	80000	106	Risk-Management	80000	Pending

2. LEFT JOIN

```
SELECT Employee_ID, First_name, Last_name, Department, Salary,
       Project_ID, Project_name, Budget, Status
FROM   Employees
LEFT JOIN Projects
ON employees.employees_ID = projects.employees_ID;
```

employee_ID	First_name	Last_name	Department	Salary	project_ID	Project_name	Budget	Status
1	John	Doe	IT	70000	101	AI Developed	100 000	Completed
2	Alice	Smith	HR	60 000	102	Employee-Traning	50 000	Ongoing
1	John	Doe	IT	70000	103	cybersecurity Audit	75 000	Pending
3	Bob	Johnson	Finance	75000	104	Financial-Analysis	90 000	Ongoing
5	Emma	Wilson	Sales	65 000	105	Market-Expansion	65 000	Completed
6	Michael	Clark	Finance	80 000	106	Risk-Management	80 000	Pending

3. RIGHT JOIN

```
SELECT project_ID, project_name, Budget, Status, Employee_ID, First_name,
       Last_name, Department, Salary
~~RIGHT FROM Projects~~
~~FROM Employees~~ FROM Projects
~~Right~~
RIGHT JOIN ~~Projects~~
ON Employee.employee_ID = projects.employee_ID;
```

3. ~~Righ~~ RIGHT JOIN

```
SELECT project_ID, project_name, Budget, Status, Employee_ID, First_name,
       Last_name, Department, Salary
FROM Projects
RIGHT JOIN Employees
ON projects.employee_ID = Employees.employee_ID;
```

Project_ID	Project_name	Budget	Status	Employee_ID	First_name	Last_name	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70 000
102	Employee_Training	50 000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity_Audit	75 000	Pending	1	John	Doe	IT	70 000
104	Financial_Analysis	90 000	Ongoing	3	Bob	Johnson	Finance	75 000
105	Market_expansion	65 000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk_management	80 000	Pending	6	Michael	Clark	Finance	80 000
				4	David	Brown	IT	72 000

4. FULL OUTER JOIN

```
SELECT Employee_ID, First_name, Last_name, Department, Salary, Project_ID,
       Project_name, Budget, Status
FROM Employees
FULL OUT JOIN Projects
ON Employees.employee_ID = projects.employee_ID;
```

Employee_ID	First_name	Last_name	Department	Salary	Project_ID	Project_name	Budget	Status
1	John	Doe	IT	70 000	101	AI Development	100 000	Completed
2	Alice	Smith	HR	60000	102	Employee Training	50 000	Ongoing
3	Bob	Johnson	Finance	75 000	104	Financial Analysis	90 000	Pending
4	David	Brown	IT	72 000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65000	Ongoing
6	Michael	Clark	Finance	80 000	106	Risk management	80000	Complete
1	John	Doe	IT	70 000	103	Cybersecurity Audit	75 000	Pending

FILTERING STATEMENT

4 SELECT Employee_ID, First_name, Last_name, Department, Salary
FROM Employees
WHERE Salary > 70 000;

Employee_ID	First_name	Last_name	Department	Salary
3	Bob	Johson	Finance	75000
4	David	Brow	IT	72000
6	Michael	Clark	Finance	80 000

8.

```
SELECT Employee_ID, First_name, Last_name, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');
```

employee_ID	First_name	Last_name	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80 000

9.

~~SELECT project~~

```
SELECT
    Project_ID,
    Project_name,
    Budget,
    Status
FROM projects
WHERE
    ~~Status NOT IN ('completed~~
    Status IN ('pending', 'Ongoing');
```

Project_ID	Project_name	Budget	Status
102	Employee_training	50 000	Ongoing
103	Cybersecurity_Audit	75 000	Pending
104	Financial_analysis	90 000	Ongoing
106	Risk_Management	80 000	Pending.

10. SELECT

Project_ID,
Project_name,
Budget,
Status

FROM Projects
WHERE Budget > 70000 AND Status! = 'Completed';

Project_id	Project_name	Budget	Status
103	Cybersecurity-Audit	75 000	Pending
	Financial_analysis	90 000	Ongoing
	Risk_management	80 000	Pending

11. SELECT
employee_ID, first_name, Last_name, Department, Salary, City
FROM Employees
WHERE City = 'New York' OR City = 'Toronto'
ORDER BY Salary DESC;

Employee_ID	First_name	last_name	Department	Salary	City
1	John	Doe	IT	70000	New York
3	Bob	Johnson	Finance	75000	Toronto
6	Michael	Clark	Finance	80000	New York

12.
SELECT
Employee_ID, first_name, Last_name, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

Employee_ID	first_name	Last_name	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

Aggregates

13.

```
SELECT
    Department,
    Sum(salary) AS Total_Salary
FROM Employees
GROUP BY Department
ORDER BY Total_~~sly~~salary DESC;
```

Department	Total_Salary
IT	142 000
Finance	155 000
HR	60 000
Sales	65000

14.

```
SELECT
    City,
    ~~Average_Salary(~~
    Ave(Salary) AS Average_Salary
FROM Employees
GROUP BY City
HAVING Avg(salary) >65 000
```

City	Average_Salary
New York	75000
Toronto	75000
LONDON	72000

15.

```
SELECT
    Department,
    COUNT(employee_ID) AS Employee_count
FROM Employees
GROUP BY department
HAVING count(employee_ID) >1;
```

Department	Employee_Count
IT	2
Finance	2

16.

```
SELECT
        status,
        ~~Project~~
        COUNT (project_ID) AS Project_count
FROM Projects
GROUP BY Status
HAVING COUNT (Project_ID) >=2;
```

Status	Project_count
Pending	2
Ongoing	2
Completed	2

17.

```
SELECT
        Employee_ID, First_name, Last_name, Total_project Budget
        SUM (Budget) AS Total_~~Budget~~ Project - Budget
FROM Projects
WHERE Projects > 150 000;
```

Employee_ID	First_name	Last_name	Total_~~Budget~~ Project - Budget
1	John	Doe	175 000